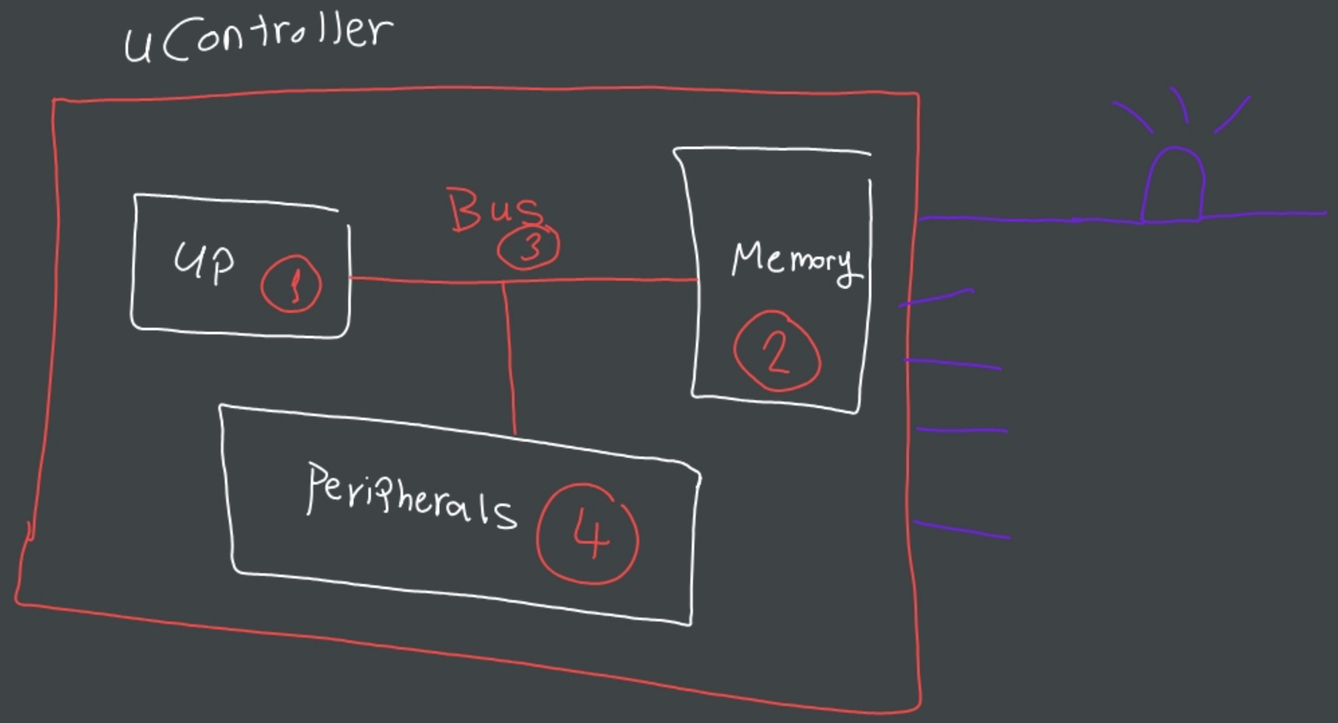
Architecture

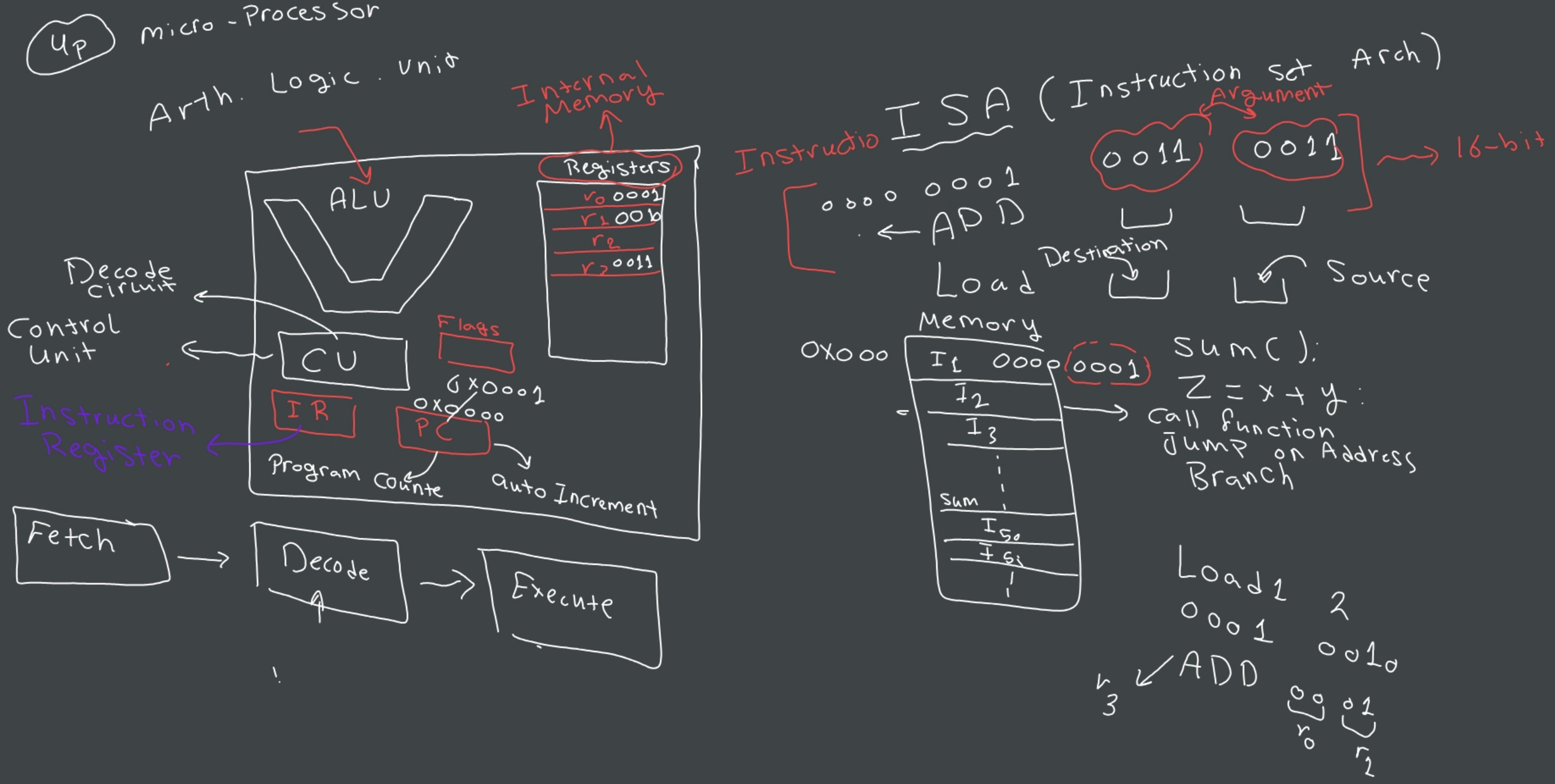


0xABC = 0x11

Printf("%x", 0xABC);

↓

0x10


µp
micro - Processor
Arth. Logic. Unit
Internal Memory
ALU
Registers
r_0 0001
r_1 001b
r_2
r_3 0011
Decode circuit
Control Unit
CU
Flags
0x0001
0x0000
IR
PC
Instruction Register
Program Counte
auto Increment
Fetch
Decode
Execute
ISA (Instruction set Arch)
Instructio
0000 0001
ADD
Load
Argument
0011
0011
16-bit
Destination
Source
Memory
0x000
I_1 0000 0001
I_2
I_3
sum
I_{50}
I_{51}
sum():
Z = x + y;
call function
Jump on Address
Branch
Load 1 2
0001 0010
ADD
r_3
00 01
r_0 r_1

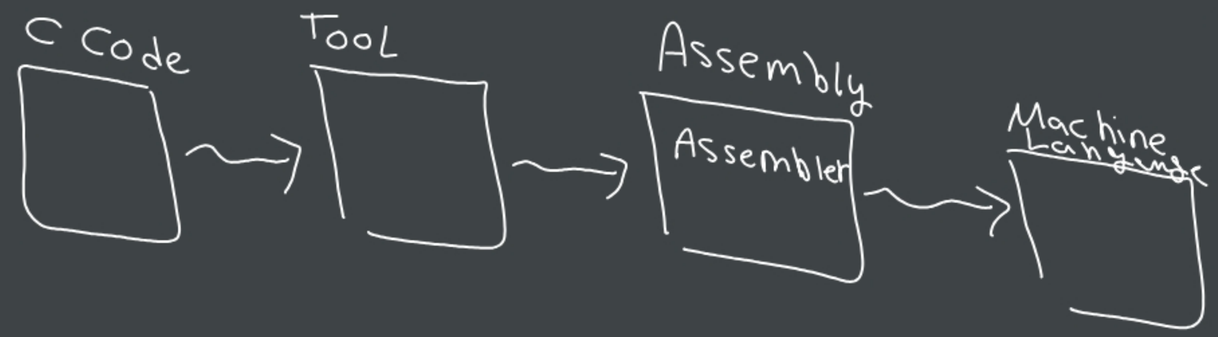
Z = X + Y ;

Load 0 X ⟶ 0 1 - - -

Load 1 Y

ADD 0 0 0 1

STORE 3 Z



8-bit Instruction Set

		Argument 4-bit
Load 0	0 0 0 0	
Load 1	0 0 0 1	
Load 2	0 0 1 0	
Load 3	0 0 1 1	
Add	0 1 0 0	2-bit , 2-bit
STORE 0		
⋮	⋮	
STORE 3		

Memory

Non-Volatile

FLASH
E^2PROM
ROM

Cost ⇝ High Voltage
⤷ Very Slow

Volatile memory

SRAM (Static RAM)
DRAM (Dynamic RAM)

- Async DRAM
- SDRAM
- DDRAM

Static RAM

Based on Transistors

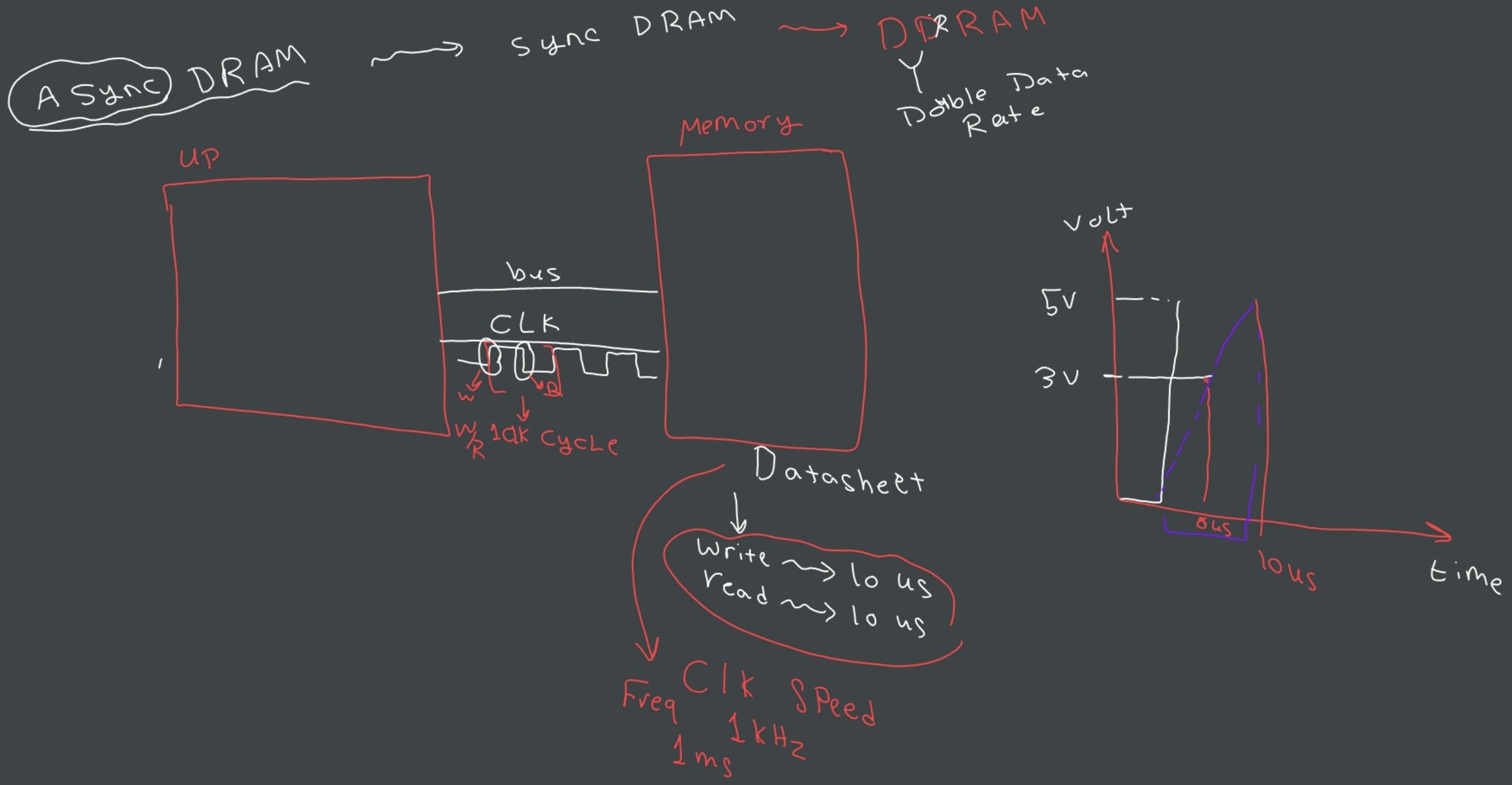
Bit → 6 Transistors

Very Fast
Very Cost ↑↑

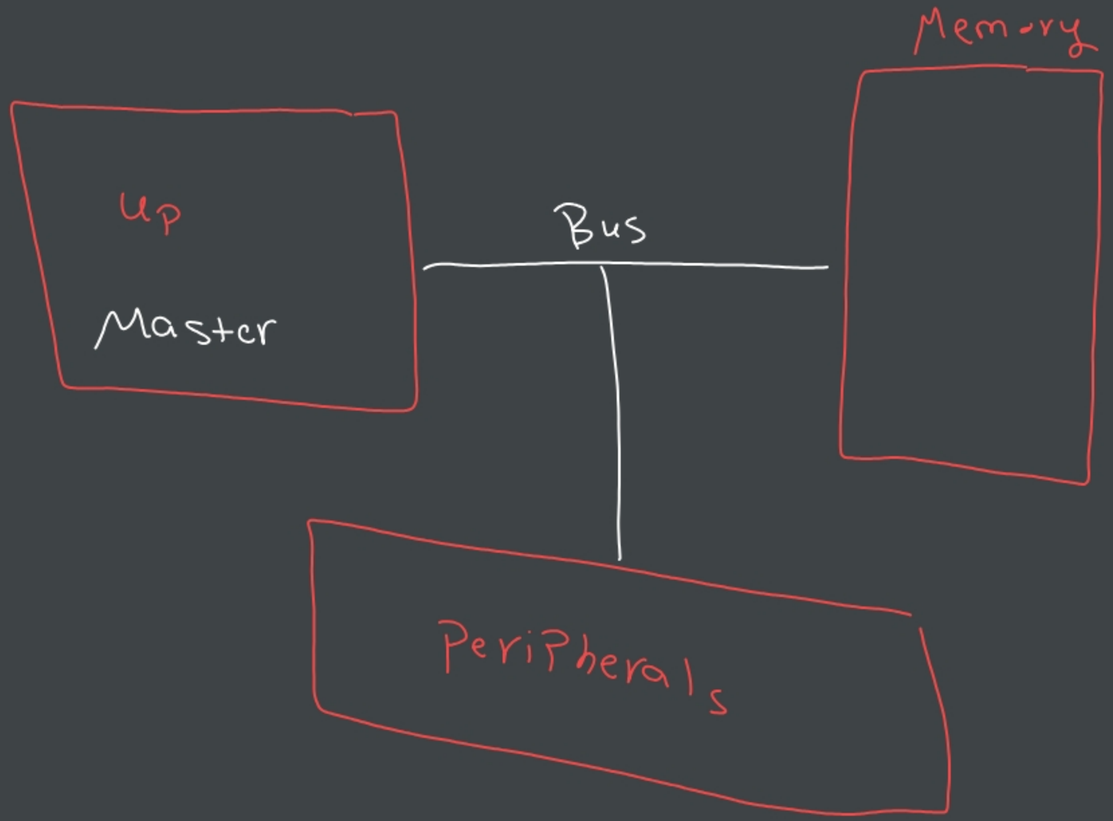
Dynamic RAM

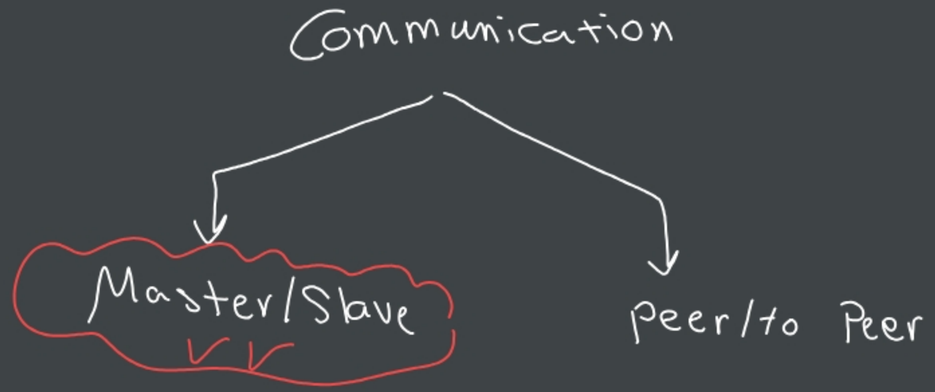
Based Capacitors

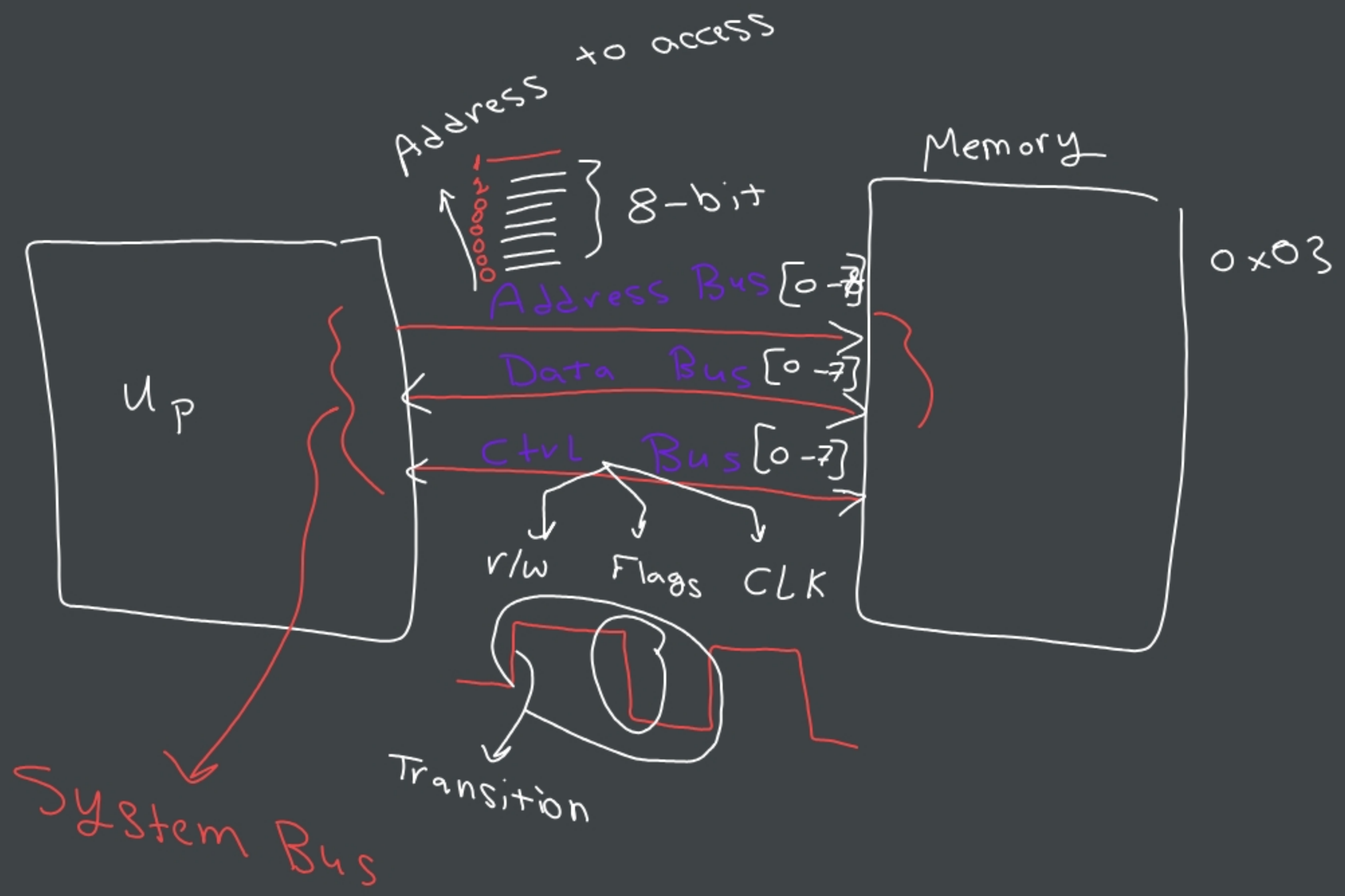
⇝ very Low Cost
⇝ Power consumption ↑
⇝ Time (very Slow)


ASync DRAM
Sync DRAM
DDR RAM
Double Data Rate
Memory
UP
bus
CLK
W
R
W/R 1clk cycle
Datasheet
Write ~> 10 us
Read ~> 10 us
Freq Clk Speed
1kHz
1ms
Volt
5V
3V
8us
10us
time

③ Bus


Memory
Up
Master
Bus
Peripherals

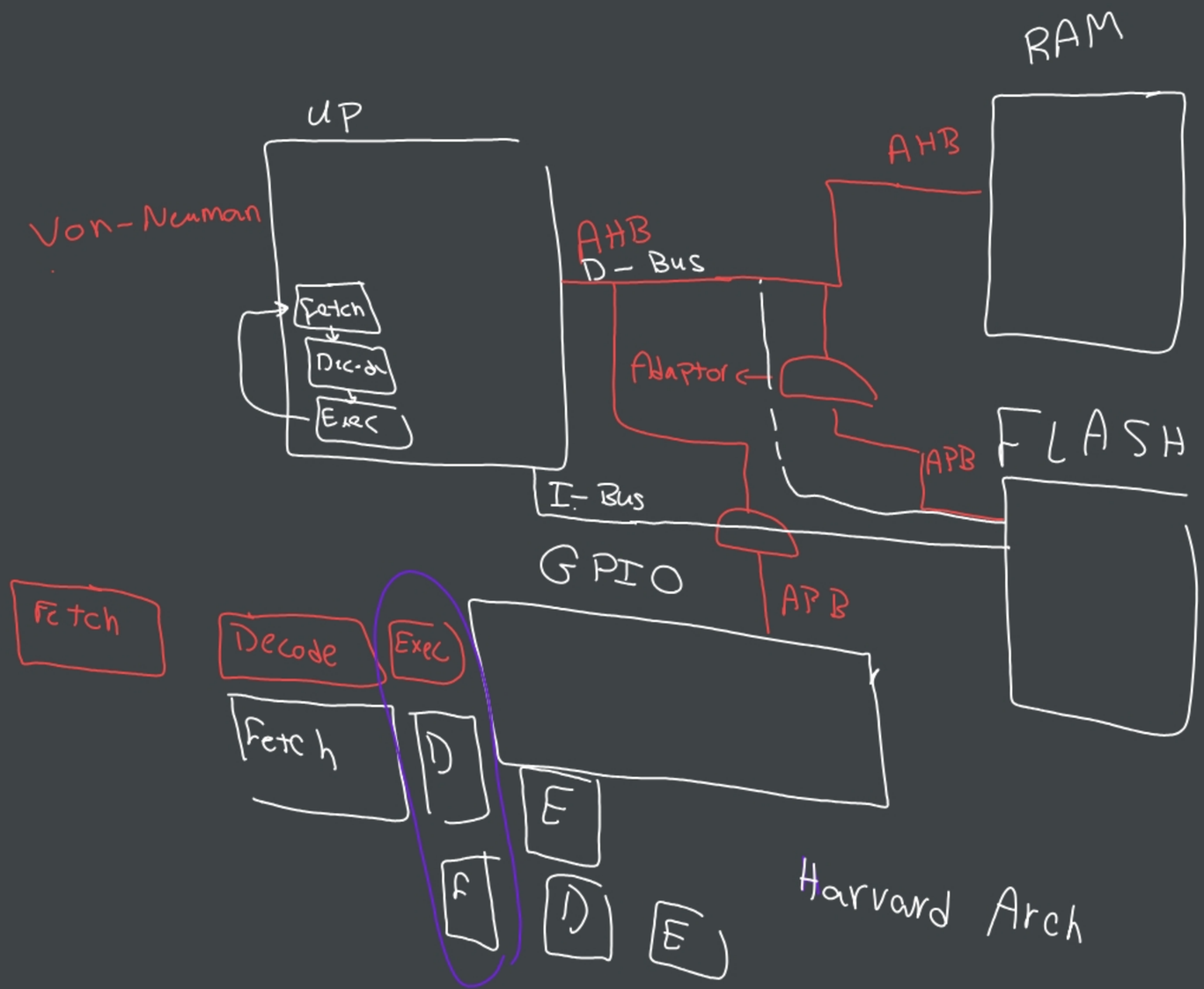

Communication
Master/Slave
Peer/to Peer


Address to access
8-bit
Memory
0x03
Up
Address Bus [0-7]
Data Bus [0-7]
Ctrl Bus [0-7]
r/w
Flags
CLK
Transition
System Bus

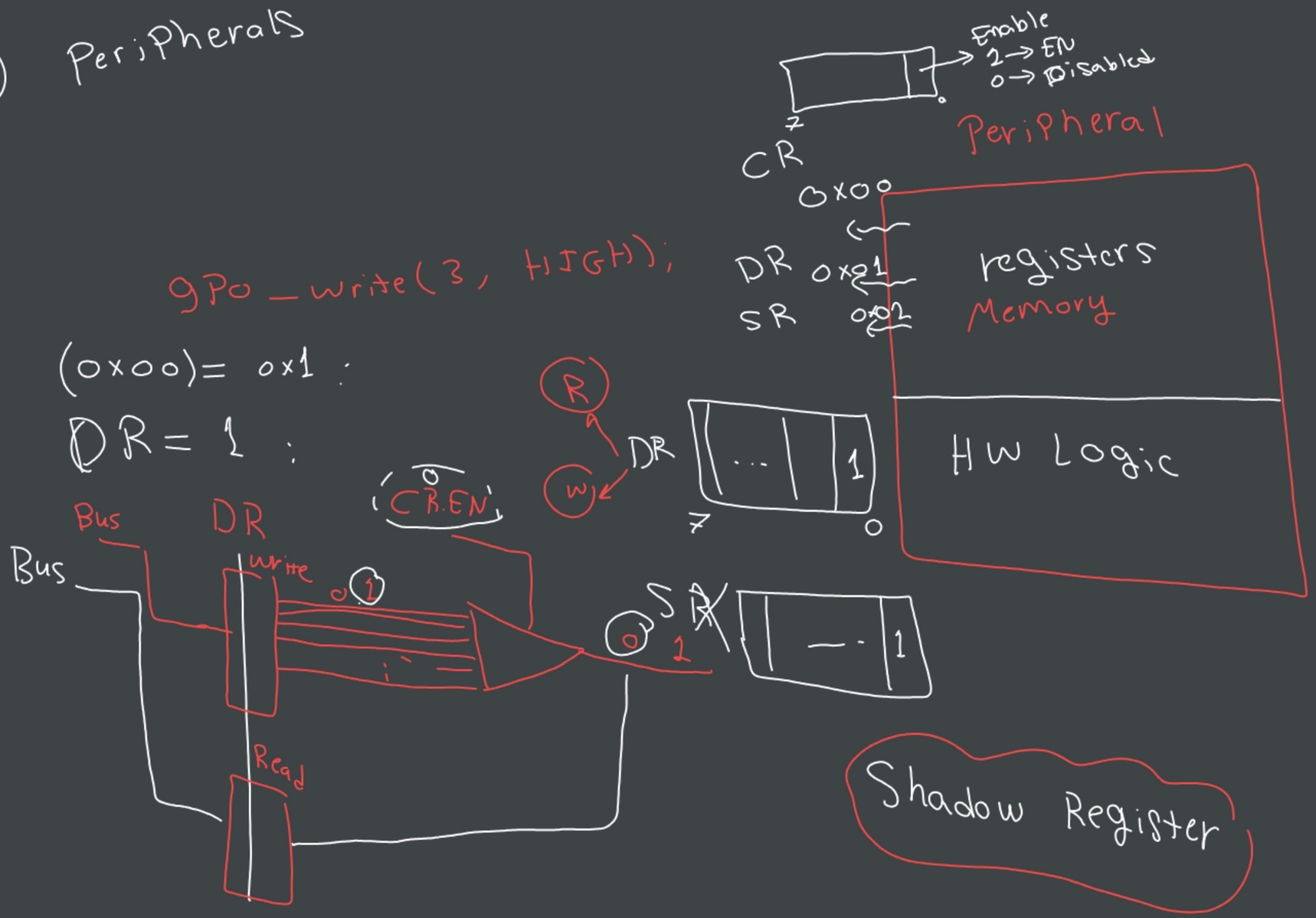
AMBA

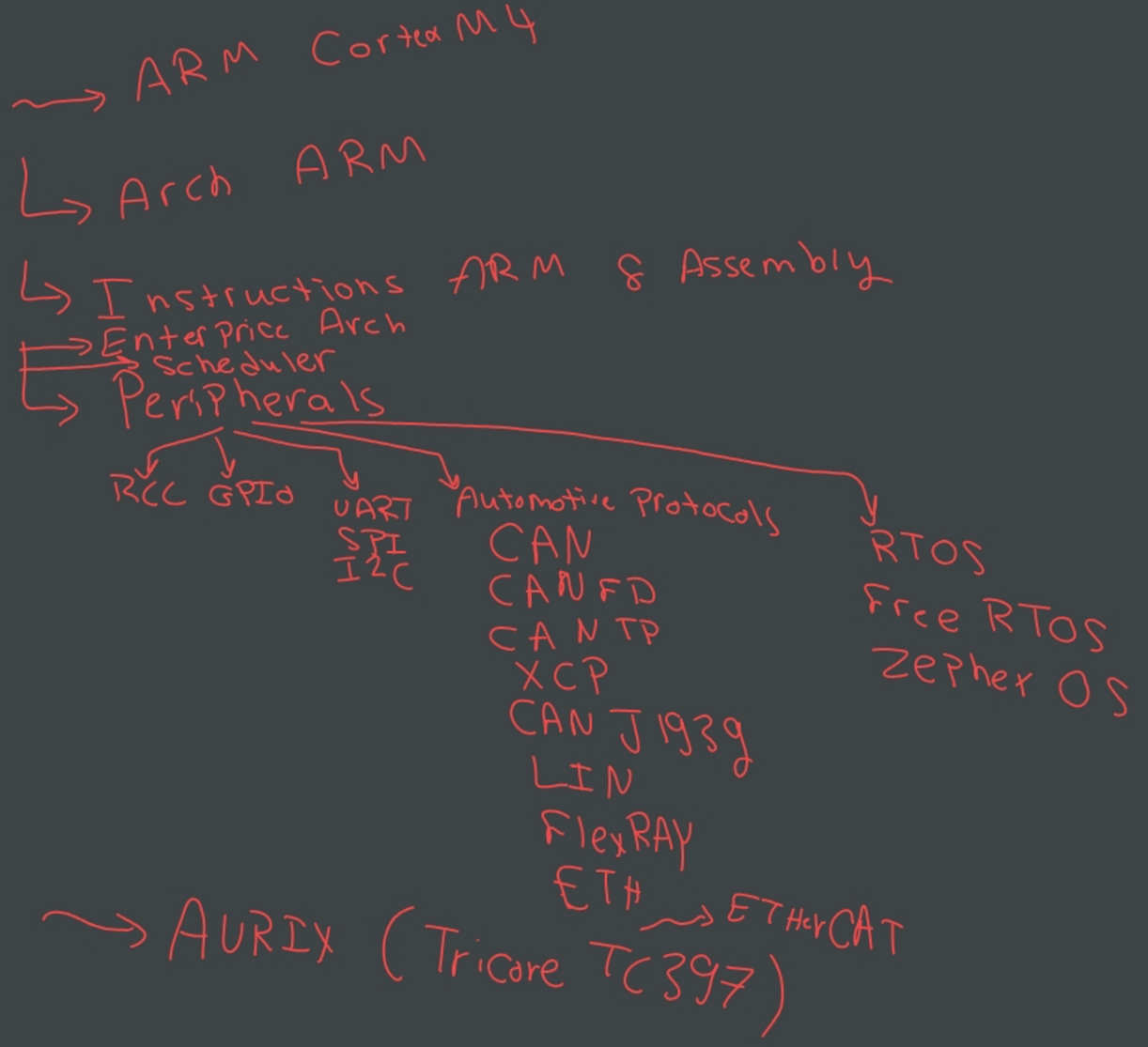
Advanced uC Bus Arch.

- → AHB ⇝ High Speed Bus
- → APB ⇝ Slow Speed [Peripherals]
- → AXI-4 ⇝ Similar to AHB (High Speed) ⇝ FPGA
- → AXI-4 Lite ⇝ Low Speed ⇝ FPGA
- → AXI-STREAM ⇝ Video Streaming ⇝ Very High Speed.


RAM
uP
Von-Neuman
AHB
AHB
D- Bus
Fetch
Dec-de
Exec
Adaptor
APB
FLASH
I- Bus
GPIO
APB
Fetch
Decode
Exec
Fetch
D
E
F
D
E
Harvard Arch

4 Peripherals




ARM Cortex M4
Arch ARM
Instructions ARM & Assembly
Enterprice Arch
Scheduler
Peripherals
RCC
GPIO
UART
SPI
I2C
Automotive Protocols
CAN
CAN FD
CAN TP
XCP
CAN J1939
LIN
FlexRAY
ETH
ETHerCAT
RTOS
Free RTOS
Zepher OS
AURIX (Tricore TC397)